

Multiple Choice (10 marks)

1. What is the maximum number of phases that can be at equilibrium with each other in a three component mixture?
 - (a) 2
 - (b) 3
 - (c) 4
 - (d) 5
2. Which of the following is an n-type semiconductor?
 - (a) Silicon
 - (b) Diamond
 - (c) Silicon carbide
 - (d) Arsenic-doped silicon
3. Considering 0.1 M aqueous solutions of each of the following, which solution has the lowest pH?
 - (a) Na_2CO_3
 - (b) Na_3PO_4
 - (c) Na_2S
 - (d) NaCl
4. The fact that the infrared absorption frequency of deuterium chloride (DCl) is shifted from that of hydrogen chloride (HCl) is due to the differences in their
 - (a) electron distribution
 - (b) dipole moment
 - (c) force constant
 - (d) reduced mass
5. What is the NMR frequency of ^{31}P in a 20.0 T magnetic field?
 - (a) 54.91 MHz
 - (b) 239.2 MHz
 - (c) 345.0 MHz
 - (d) 2167 MHz

True / False (10 marks)

Answer True or False

6. True or False: Each hybrid sp^3 orbital can hold a maximum of two electrons, not four.
7. True or False: The reduction of D-xylose with NaBH_4 produces a racemic mixture of enantiomers.
8. True or False: Calcium (Ca) has the lowest electron affinity among the elements listed due to its larger atomic size and lower effective nuclear charge.
9. True or False: The thermal stability of group-14 hydrides increases in the order: $\text{CH}_4 < \text{SiH}_4 < \text{GeH}_4 < \text{SnH}_4 < \text{PbH}_4$.
10. True or False: The polarization of a proton is 1.148×10^{-6} at 0.335T and 3.598×10^{-5} at 10.5T .

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the Heisenberg uncertainty principle applies to a quantum mechanical particle in the lowest energy level of a one-dimensional box, focusing on the implications for position and momentum.
12. Discuss the decay of a radioactive isotope used in diagnostic imaging with a half-life of 6.0 hours. Calculate the remaining activity 24 hours post-delivery from an initial activity of 150 mCi, and explain the underlying principles of radioactive decay in your analysis.

End of Paper